

Type of product	Total samples	Total no.* of o-c residues	Total samples	Total no.* of o-c residues	Total samples	Total no.* of o-c residues
Leaf veg.	159	34	131	20	141	40
Root veg. (including potatoes)	144	36	98	18	129	24
Pip fruit	104	139	25	45	60	39
Stone fruit	36	36	24	30	38	25
Citrus fruit	12	4	6	3	58	17

* Some samples have two or more organochlorine residues present.

TABLE 4.—Pesticides present in excess of New Zealand tolerances and produce involved

Compound	Number over tolerance			Commodities involved
	1971	1972	1973	
Ornethoate	-	2	-	Pip and stone fruits
TDE	1	2	-	Citrus and stone fruits
DDT	1	-	1	Leaf and root vegetables
Dimethoate	1	-	-	Citrus fruit
Dicofol	1	-	-	Leaf vegetables
Parathion methyl	-	1	2	Citrus fruit and root vegetables
Mevinphos	-	1	1	Berry fruit
Diazinon	-	-	1	Citrus fruit
Total number of samples above tolerance	4	6	5	
Total number of samples analysed in year	159	191	174	

safe to have residues at a particular level in all members of a class of food items of the diet, and it is assumed that a normal mixed diet can safely accommodate the occasional food item with a high residue.

The present monitoring programme on food crops has shown that few abuses occur in the use of pesticides. Many of the high residues detected were only slightly in excess of the regulation limit, and probably reflect variations in climate and application technique.

ACKNOWLEDGMENTS

We thank the New Zealand Department of Health and its Inspectors of Health for the organisation and collection of samples.

REFERENCES

- BREWERTON, H. V., MCGRATH, H. J. W. 1967. Insecticides in human fat in New Zealand. *N.Z. Journal of Science* 10: 486-92.
- CORLESTON, J. F., HUNNEGO, J. N., HARRISON, D. L. 1973. Organochlorine insecticide levels in adult New Zealanders—a five year study. *N.Z. Journal of Science* 16: 27-39.
- ELGAR, K. E. 1971. The identification of pesticides at residue concentrations. *Advances in Chemistry Series* 104 (Ed. R. E. Gould) American Chemical Society, Washington D.C.
- HUGHES, J. T. 1972. Spray residues on fruit and vegetables. *N.Z. Journal of Agriculture* 125(6): 60-63.
- MENDOZA, C. F., WALES, P. J., MCLEOD, H. A., MCKINLEY, W. P. 1968. Enzymatic detection of ten organophosphorus pesticides and carbaryl on thin-layer chromatograms: an evaluation of indoxyl, substituted indoxyl and 1-naphthyl acetates as substrates of esterases. *Analyst* 93: 34-38.
- NEW ZEALAND DEPARTMENT OF AGRICULTURE. 1970. *Annual report of the Research Division 1969-70. STATUTORY REGULATIONS*. 1973a. Food and Drug Regulations 1973/79.
- 1973b. Food Additives Notice 1973/276.

Polychlorinated biphenyls and organochlorine pesticides in human fat in New Zealand

S. R. B. SOLLY and V. SHANKS

Wallaceville Animal Research Centre, Research Division,
Ministry of Agriculture and Fisheries, Upper Hut, New Zealand

Organochlorine residues were determined in 51 human fat samples obtained at autopsy in one New Zealand city in 1973. DDT, DDE, hexachlorobenzene, and polychlorinated biphenyls (PCBs) were present in all samples at mean levels of 0.70 and 5.64, 0.31, and 0.82 ppm respectively. Dieldrin (mean level 0.21 ppm), β -BHC, TDE, and lindane occurred in measurable amounts in 96%, 37%, 29%, and 22% of the samples respectively. DDE constituted 88.7% of the total DDT residue.

Residues of DDT and PCB increased with age, whilst residues of lindane decreased with age ($P < 0.05$, $P < 0.01$ and $P < 0.01$ respectively).

Residues of heptachlorobiphenyls were significantly higher in males than in females ($P < 0.05$). Significant differences between sexes were not observed for other organochlorine compounds.

Gas chromatograms of human fat extracts were qualitatively similar to chromatograms obtained from extracts of pig and beef fat environmentally contaminated with PCBs.

INTRODUCTION

Polychlorinated biphenyls (PCBs) comprise a range of industrial compounds, first produced commercially in 1929 and subsequently used in electrical equipment, paints, heat transfer fluids, plastics, and a variety of other products. They are mixtures incorporating about 50 of the 210 compounds theoretically possible from the chlorination of biphenyl, and they are produced with differing chlorine contents according to their intended use; the most widely used industrial mixtures average between 40% and 60% chlorine content.

They are stable fat-soluble materials which can accumulate in food chains (Hammond 1972). Organochlorines, including PCBs, may now be the most abundant synthetic pollutants in the global environment (Risbrough *et al.* 1968). In September 1972 an inter-departmental 'task force' was set up in the U.S.A. to "co-ordinate a government-wide investigation into PCB contamination of food and other products". This taskforce recommended (U.S. Interdepartmental Task Force on PCBs 1972) restriction of PCBs to essential or irreplaceable uses involving minimal direct human exposure so as to minimise further introduction of these compounds into the food supply and the environment. To our knowledge PCBs in New Zealand have been voluntarily restricted by the proprietors of the product to uses that comply with the U.S. recommendations.

Some studies of the occurrence of PCBs in human adipose tissue have been made. For example Price & Welch (1972) found 81% of 194 samples from the general population of U.S.A. contained detectable amounts of PCBs in concentrations up to 2 ppm, and 9% contained concentrations greater than 2 ppm.

Received 5 June 1974.

Ministry of Agriculture & Fisheries, Research Division, New Zealand.
Wallaceville Animal Research Centre Publication No. 428.

TABLE 2.—Mean levels (ppm, fat basis) and standard deviations for 8 analyses for organochlorine pesticides and for 7 analyses for polychlorinated biphenyls in human adipose tissue sample No. 1

Compound	Mean	Standard deviation
HCB	0.162	0.007
α -BHC	<0.01	—
β -BHC	<0.20	—
γ -BHC (lindane)	<0.05	—
DDE	3.38	0.12
DDT	0.535	0.022*
<i>op'</i> -DDT	<0.02	—
TDE	<0.02	—
Dieldrin	0.127	0.005
PCB 4	0.07	0.01
PCB 5	0.15	0.03
PCB 6	0.14	0.01
PCB 7	0.07	0.00
PCB 8	0.01	0.00
PCB total	0.44	0.04

* 7 analyses.

TABLE 3.—Levels of organochlorines (ppm, extracted fat basis) in human adipose tissue

Compound	Male (M)		Female (F)		Overall (M + F)	
	Mean	Range	Mean	Range	Mean	Slope
DDT*	0.70	0.12–1.84	0.70	0.13–2.11	0.70	$P < 0.05$
DDE	5.97	2.50–15.88	5.18	1.55–27.14	5.64	0.0039
TDE	0.07	<0.02–0.13	0.07	<0.02–0.14	0.07	n.s.
Total DDT	6.68	2.62–17.85	5.91	1.68–29.39	6.36	n.s.
Dieldrin	0.21	<0.02–1.96	0.21	<0.02–0.64	0.21	n.s.
HCB	0.34	0.15–1.25	0.28	0.04–0.80	0.31	n.s.
β -BHC	0.47	<0.20–4.05	0.28	<0.20–0.66	0.39	n.s.
Lindane*	0.12	<0.05–0.44	0.07	<0.05–0.22	0.10	$P < 0.01$
PCB 4	0.13	0.01–0.93	0.11	0.05–0.25	0.12	n.s.
PCB 5	0.25	0.13–1.49	0.21	0.07–0.53	0.24	n.s.
PCB 6	0.31	0.18–1.07	0.22	0.09–0.91	0.28	n.s.
PCB 7	0.17†	0.08–0.94	0.10†	0.04–0.41	0.14	n.s.
PCB 8	0.04	0.01–0.24	0.03	0.01–0.11	0.04	n.s.
PCB total (sum of PCB 4–8)	0.90	0.50–4.67	0.67	0.29–2.09	0.81	n.s.
PCB total* (measured as DCB)	0.89	0.48–3.80	0.76	0.33–2.08	0.83	$P < 0.01$
						0.0058

* Results for these compounds have been adjusted to the overall mean age

† Significant difference ($P < 0.05$)

n.s. No significant correlation

typical chromatograms of PCB-contaminated beef and pig fat are illustrated in Fig. 1a and 1b. The profiles of the chromatograms obtained from human fat extracts, Fig. 2b, are qualitatively similar to those from extracts of animal fats and to those from an injection of Aroclor 1262 (Fig. 2a), although peaks that occur between DDE and DDT in the Aroclor chromatogram are absent from the pig fat and human fat chromatograms. These peaks

